MCU: Bottoming but Not Out

April 17, 2026 08:25 AM GMT

The recovery looks more L-shaped unless we see a meaningful decline in memory prices or a pickup in edge AI.

Key Takeaways

- The April MCU spot market pricing hike suggests the cycle is bottoming, though pricing may remain volatile in the coming months.
- Contract prices are now only reflecting foundry/packaging cost increases, suggesting limited GM recovery.
- Component costs and availability remain the key differentiators among companies.
- We prefer large MCU players (e.g., GigaDevice) and WiFi MCU vendors (Espressif) for LT edge AI opportunities.

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Strong spot price hike in April; the base is already low: In April, STM's 32-bit MCU was up 25% MoM to Rmb13.7, while GigaDevice's 32-bit MCU spot price rose 53% MoM to Rmb9.5, catching up the rally in STM's 32bit MCU pricing. In late March, STMicro further raised prices following its pre-CNY increase. Besides CMSemicon (688380.SS, NC), smaller companies such as Puya (688766.SS, NC) and NSING Technologies (300077.SZ, NC) also issued price hike notices.

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But limited impact on contract pricing and gross margin: Greater China players are trying to raise contract prices. For example, both Gigadevice and Nuvoton have raised prices for some low margin MCU products. That said, this does not signal a broad recovery. Supply was indeed tight for smaller MCU design houses. However, most MCUs are produced on 40nm/55nm nodes using 12-inch capacity, which remains less constrained than 8-inch capacity. We therefore do not expect meaningful price increases from the big MCU companies. Any price hike is likely to be a pass-through of higher costs, suggesting limited GM expansion.

Demand remains challenged; memory sourcing becomes the key: Consumer demand in April was slightly weaker as seasonal 'back to work' and 'back to school' demand faded, while the recovery in auto and industrial demand remains slow. At the same time, we see access to reasonably priced memory as a key differentiator among MCU vendors. For example, Nuvoton could expand MPU share through bundling with Winbond's DDR4, similar to GigaDevice. Espressif's ESP32 is seeing higher demand given its lower memory content.

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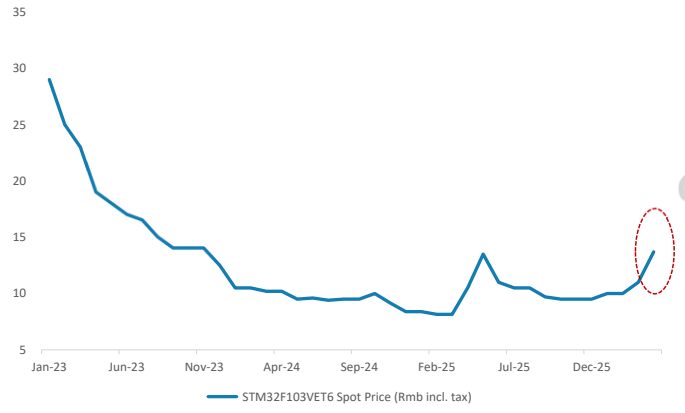
Stock implications - awaiting a clearer turnaround: We remain OW on Espressif but revise down our earnings estimates and PT. We like the company's RISC V advantage, ecosystem breadth, and recent OPEN Claw pickup, but weak consumer demand may create downside risk to earnings. We remain UW on Nuvoton but raise our PT on MPU share expansion and BMC/BMIC business opportunities. We are awaiting GM recovery before turning more positive on the stock.

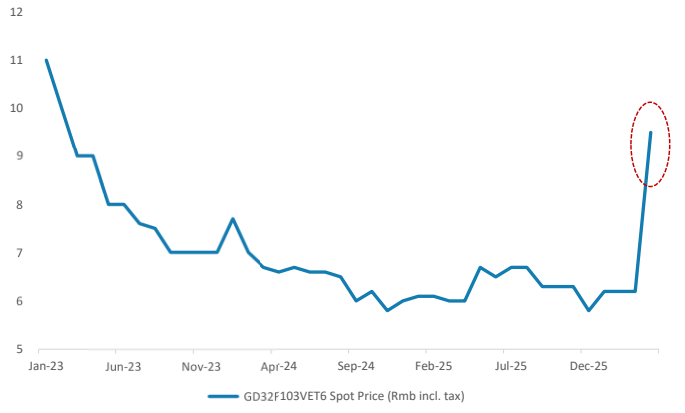
Key Charts

Exhibit 1: STM32F103VET6 spot market price trend



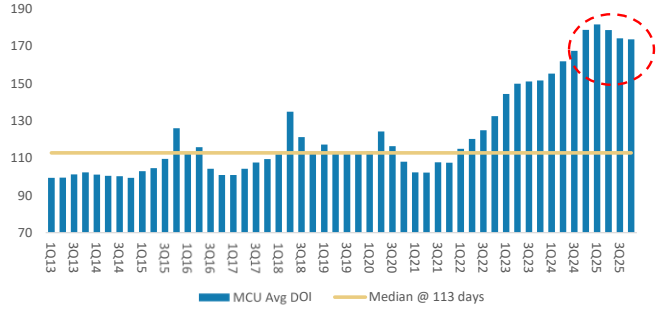
Source: Bom.ai, Morgan Stanley Research

Exhibit 2: GD32F103VET6 spot market price trend



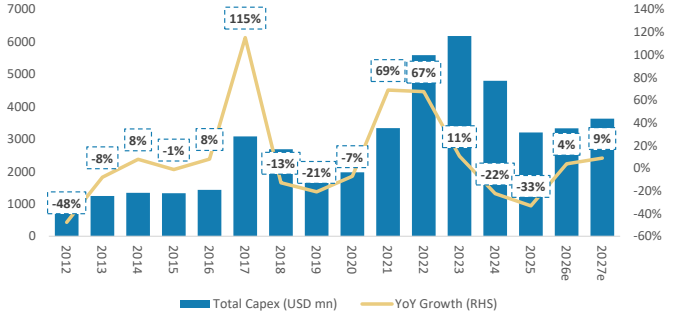
Source: Bom.ai, Morgan Stanley Research

Exhibit 3: Inventory days for producers peaked in 1Q25 and was flat in 4Q25



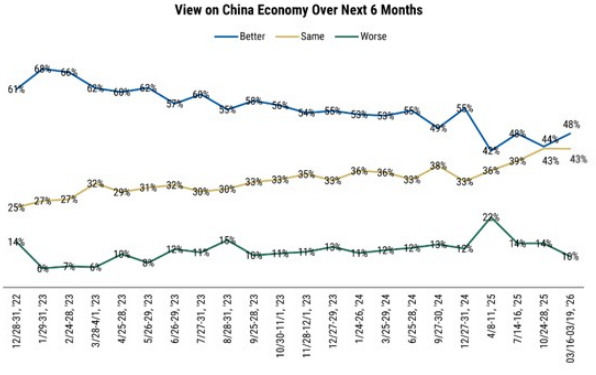
Source: Company data, Bloomberg, Morgan Stanley Research

Exhibit 4: Leading MCU companies' capex declined for 2 years and is showing slight growth in 2026



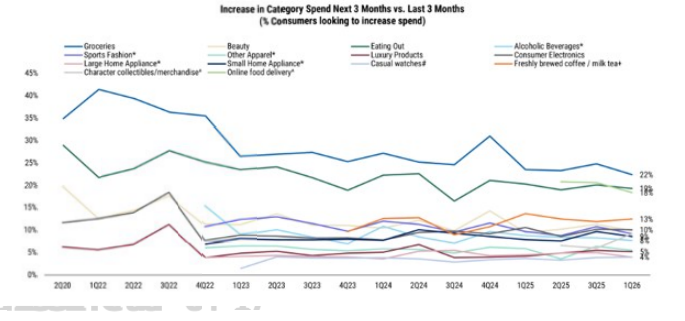
Source: Company data, Bloomberg (e) estimates, Morgan Stanley Research.

Exhibit 5: Consumer sentiment shows mild improvement, though caution remains



Source: AlphaWise, Morgan Stanley Research. Note: survey date as of March 16-19, 2026.

Exhibit 6: Consumer electronics' spending intention was stable recently



Source: AlphaWise, Morgan Stanley Research. Note: survey date as of March 16-19, 2026.

Espressif: Earnings Estimate Revisions

We cut our 2026/27 EPS estimates by 12%/11% and introduce our 2028 forecast: We factor in actual 2025 results and cut our revenue forecasts, as we believe overall consumer electronics demand is being affected by higher memory prices. We have also raised our R&D expense assumptions, as the company guided to a 20% increase in R&D expense for 2026. As a result, our 2026/27 EPS estimates are lowered by 12%/11%. We also introduce our 2028 forecast.

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Exhibit 7: Espressif: Estimate revisions

Rmb mn	New 2026E	Previous 2026E	Diff.	New 2027E	Previous 2027E	Diff.	New 2028E
Net sales	3,052	3,069	-1%	3,804	4,001	-5%	4,552
COGS	1,639	1,715		2,106	2,290		2,554
Gross profit	1,413	1,354	4%	1,697	1,711	-1%	1,999
Operating expenses	910	755		1,059	971		1,262
Operating profit	503	600	-16%	639	739	-14%	737
Non-op. income (exp.)	60	28		60	28		60
Pretax Income	563	628	-10%	699	767	-9%	797
Taxes	28	13		35	15		40
Net income	538	611	-12%	667	748	-11%	760
Reported EPS (Rmb)	3.43	3.90	**-12%**	4.26	4.77	**-11%**	4.85
Margins							
Gross margin	46.3%	44.1%	2 ppt	44.6%	42.8%	2 ppt	43.9%
Operating margin	16.5%	19.5%	-3 ppt	16.8%	18.5%	-2 ppt	16.2%
Pretax margin	18.4%	20.4%	-2 ppt	18.4%	19.2%	-1 ppt	17.5%
Net margin	17.6%	19.9%	-2 ppt	17.5%	18.7%	-1 ppt	16.7%

Source: Morgan Stanley Research (E) estimates

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Exhibit 8: Espressif: Quarterly financial statement

Rmb mn(YE DEC31)	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Total Revenues	**558**	**688**	**667**	**653**	**659**	**716**	**820**	**858**	**810**	**894**	**1,026**	**1,074**	**2,565**	**3,052**	**3,804**	**4,552**
Sequential Change	*2.0%*	*23.3%*	*-3.1%*	*-2.1%*	*0.9%*	*8.6%*	*14.6%*	*4.7%*	*-5.6%*	*10.3%*	*14.8%*	*4.7%*				
Change vs Year Ago	*44.1%*	*29.0%*	*23.5%*	*19.4%*	*18.1%*	*4.1%*	*23.0%*	*31.4%*	*23.0%*	*24.9%*	*25.2%*	*25.2%*	*27.8%*	*19.0%*	*24.6%*	*19.7%*
Cost of Sales	316	367	347	340	354	384	440	461	449	495	568	594	1,369	1,639	2,106	2,554
Percent of Revenues	*57%*	*53%*	*52%*	*52%*	*54%*	*54%*	*54%*	*54%*	*55%*	*55%*	*55%*	*55%*	*53%*	*54%*	*55%*	*56%*
Gross Profit	**242**	**321**	**320**	**313**	**305**	**331**	**380**	**397**	**361**	**399**	**458**	**480**	**1,196**	**1,413**	**1,697**	**1,999**
Percent of Revenues	*43.4%*	*46.7%*	*48.0%*	*48.0%*	*46.3%*	*46.3%*	*46.3%*	*46.3%*	*44.5%*	*44.6%*	*44.7%*	*44.7%*	*46.6%*	*46.3%*	*44.6%*	*43.9%*
Incremental Margin	*-131%*	*61%*	*NM*	*NM*	*-147%*	*47%*	*46%*	*46%*	*NM*	*45%*	*45%*	*44%*	*56%*	*45%*	*38%*	*40%*
Total Opex	161	183	201	229	197	214	244	256	227	249	285	298	774	910	1,059	1,262
Percent of Revenues	*28.9%*	*26.6%*	*30.1%*	*35.1%*	*29.8%*	*29.9%*	*29.7%*	*29.8%*	*28.0%*	*27.9%*	*27.7%*	*27.8%*	*30.2%*	*29.8%*	*27.8%*	*27.7%*
R&D	126	142	155	180	157	170	195	204	178	197	226	236	603	726	837	1,002
Percent of Revenues	*22.6%*	*20.7%*	*23.3%*	*27.6%*	*23.8%*	*23.8%*	*23.8%*	*23.8%*	*22.0%*	*22.0%*	*22.0%*	*22.0%*	*23.5%*	*23.8%*	*22.0%*	*22.0%*
General & administrative	18	20	24	27	20	22	24	26	24	26	28	30	89	92	108	124
Percent of Revenues	*3.2%*	*2.9%*	*3.6%*	*4.1%*	*3.0%*	*3.1%*	*2.9%*	*3.0%*	*3.0%*	*2.9%*	*2.7%*	*2.8%*	*3.5%*	*3.0%*	*2.8%*	*2.7%*
Selling & marketing	17	21	21	23	20	21	25	26	24	27	31	32	82	92	114	137
Percent of Revenues	*3.1%*	*3.0%*	*3.2%*	*3.5%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.2%*	*3.0%*	*3.0%*	*3.0%*
Operating Income	**81**	**138**	**119**	**84**	**108**	**118**	**136**	**141**	**134**	**149**	**174**	**181**	**422**	**503**	**639**	**737**
Percent of Revenues	*14.5%*	*20.1%*	*17.9%*	*12.8%*	*16.4%*	*16.4%*	*16.6%*	*16.4%*	*16.6%*	*16.7%*	*16.9%*	*16.9%*	*16.5%*	*16.5%*	*16.8%*	*16.2%*
Change vs Year Ago	*162.6%*	*58.6%*	*66.2%*	*21.4%*	*33.9%*	*-14.9%*	*13.9%*	*68.3%*	*24.0%*	*26.9%*	*28.0%*	*28.4%*	*63%*	*19%*	*27%*	*15%*
Total Non-operating Income(Loss)	12	30	10	34	15	15	15	15	15	15	15	15	86	60	60	60
Interest income	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Interest expense	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0
Investment Income(loss)	5	4	3	5	15	15	15	15	15	15	15	15	17	60	60	60
Tax Rate	*-2.1%*	*-0.2%*	*9.7%*	*-2.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*1.6%*	*5.0%*	*5.0%*	*5.0%*
Net Income, Cont Ops	94	169	116	120	117	126	143	148	142	156	179	186	500	535	664	757
Percent of Revenues	*17%*	*25%*	*17%*	*18%*	*18%*	*18%*	*17%*	*17%*	*18%*	*17%*	*17%*	*17%*	*19%*	*18%*	*17%*	*17%*
Minority interest	(1)	(1)	(1)	1	1	1	1	1	1	1	1	1	(2)	3	3	3
Reported Income	**94**	**168**	**116**	**121**	**118**	**127**	**144**	**149**	**143**	**157**	**180**	**187**	**498**	**538**	**667**	**760**
Percent of Revenues	*17%*	*24%*	*17%*	*19%*	*18%*	*18%*	*18%*	*17%*	*18%*	*18%*	*18%*	*17%*	*19%*	*18%*	*18%*	*17%*
Reported EPS (Rmb)	**0.60**	**1.07**	**0.74**	**0.77**	**0.75**	**0.81**	**0.92**	**0.95**	**0.91**	**1.00**	**1.15**	**1.19**	**3.18**	**3.43**	**4.26**	**4.85**
Change vs Year Ago	*74%*	*71%*	*16%*	*37%*	*26%*	*-24%*	*25%*	*23%*	*21%*	*24%*	*25%*	*25%*	*47%*	*8%*	*24%*	*14%*

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Source: Company data, Morgan Stanley Research (E) estimates

Valuation Methodology

We cut our price target to Rmb175 from Rmb188: This reflects our lower earnings estimates for 2026-27. Our price target is our base case scenario value, derived from a residual income model. Our key assumptions are unchanged: an 8.9% cost of equity (beta of 1.1, risk-free rate of 2.0%, and risk premium of 6.0%), a payout ratio of 40%, a medium-term growth rate of 18.5%, and a terminal growth rate of 4.5%.

Accordingly, we lower our bull case value to Rmb202 from Rmb217 and our bear case value to Rmb75 from Rmb81.

Exhibit 9: Espressif: Residual income model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	**4,771**	**5,053**	**5,428**	**6,005**	**6,688**	**7,497**	**8,456**	**9,593**	**10,940**	**12,536**	**14,427**	**16,668**
Net Profit	**538**	**667**	**760**	**901**	**1,067**	**1,265**	**1,499**	**1,776**	**2,104**	**2,494**	**2,955**	**3,502**
ROAE	**11.6%**	**13.6%**	**14.5%**	**15.8%**	**16.8%**	**17.8%**	**18.8%**	**19.7%**	**20.5%**	**21.2%**	**21.9%**	**22.5%**
Residual Income	123	224	284	373	477	599	743	913	1,114	1,353	1,635	1,968
Spread	**2.7%**	**4.7%**	**5.6%**	**6.9%**	**7.9%**	**9.0%**	**9.9%**	**10.8%**	**11.6%**	**12.4%**	**13.0%**	**13.6%**
Ending Equity Capital	4,771											
PV of Forecast Period	4,305											
PV of Continuing Value	18,414											
Equity Value	**27,490**											
No. of Shares	167											
Projected Price (Rmb)	**175**											

Source: Morgan Stanley Research (E) estimates

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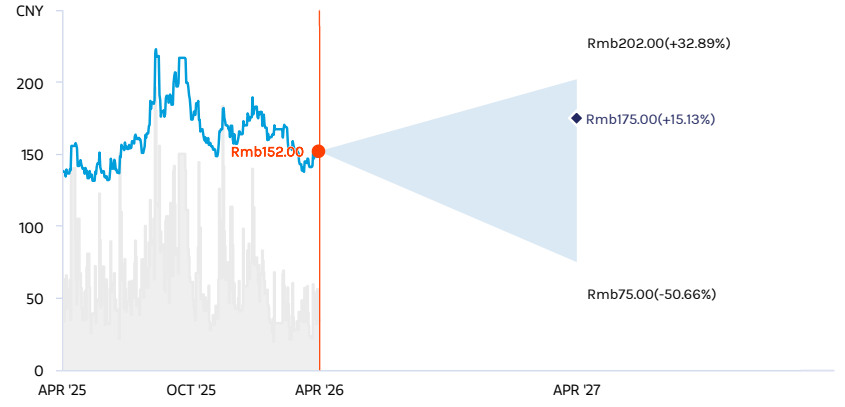
Risk Reward – Espressif Systems (688018.SS)

Edge AI growth confidence; OW

PRICE TARGET Rmb175.00

Base case, residual income model. We assume an 8.9% cost of equity (beta 1.1, risk-free rate 2.0% and risk premium 6.0%), a payout ratio of 40%, a medium-term growth rate of 18.5%, and a terminal growth rate of 4.5%.

RISK REWARD CHART



Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We believe Espressif can benefit from demand created by emerging edge AI opportunities.
- We expect the company to continue to gain market share because of a more comprehensive product portfolio, better cost performance ratio, and potentially higher RISC-V adoption rate.
- Margin has stabilized at 40%+ and we expect it to remain stable.
- 2027e P/E of 36x is below the stock's historical average of 43x since 2022; we see this as attractive, as rising ROE provides solid support for valuation.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb202.00

47x base case 2027e EPS

We assume 2025-28e revenue CAGRs of 30%+ for Wi-Fi MCUs and 40%+ for Wi-Fi modules. Gross margin of 45%+ in 2026-28e. Espressif's products are adopted in edge AI killer applications in 2026-28e.

BASE CASE Rmb175.00

41x base case 2027e EPS

We assume 2025-28e revenue CAGRs of 21% for Wi-Fi MCU and Wi-Fi modules. Gross margin of ~45% in 2026-28e. Espressif's products are adopted in more edge AI applications in 2026-28e.

BEAR CASE Rmb75.00

18x base case 2027e EPS

We assume 2025-28e revenue CAGRs of <10% for Wi-Fi MCUs and 15% for Wi-Fi modules. Gross margin below 35% in 2026-28e. No Espressif products are adopted in edge AI applications in 2026-28e.

Risk Reward – Espressif Systems (688018.SS)

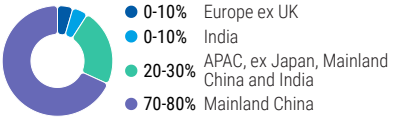
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
MCU chip shipment growth (%)	19.2	17.2	28.8	23.0
MCU chip pricing growth (%)	5.0	1.3	(2.6)	(2.7)
Gross margin (%)	46.6	46.3	44.6	43.9

INVESTMENT DRIVERS

- Wi-FI MCU unit shipments
- Pricing trends
- IoT market growth
- Wi-Fi 6 unit shipments

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- China's MCU localization goes faster than expected.
- Gains traction from new customers.
- Margin expansion is more significant.

RISKS TO DOWNSIDE

- China's MCU localization pace is slower than expected.
- Traction from new customers is less than expected.
- Margins erode amid more intense competition.

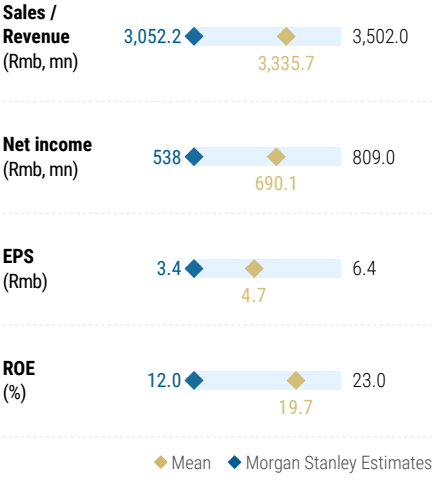
OWNERSHIP POSITIONING

Inst. Owners, % Active 87.5%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Espressif: Financial Summary

Exhibit 10: Espressif: Financial Summary

Income Statement Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	2,565	3,052	3,804	4,552
COGS	(1,369)	(1,639)	(2,106)	(2,554)
Gross profit	1,196	1,413	1,697	1,999
Operating expenses	(774)	(910)	(1,059)	(1,262)
Operating income	**422**	**503**	**639**	**737**
Non-operating income	86	60	60	60
Pre-tax income	508	563	699	797
Income tax	8	28	35	40
Minority interests	(2)	3	3	3
Reported net income	**498**	**538**	**667**	**760**
Adj.wtd.avg.shrs(m)	157	157	157	157
MW EPS (Rmb)	3.18	3.43	4.26	4.85
Reported EPS (Rmb)	**3.18**	**3.43**	**4.26**	**4.85**

Balance Sheet Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	1,025	1,223	1,269	1,414
Mkt Securities	876	876	876	876
AR/NR	364	429	534	639
Inventory	628	666	856	1,038
Other	1,043	1,043	1,043	1,043
Current Assets	**3,935**	**4,237**	**4,579**	**5,011**
Long-term investments	252	252	252	252
Fixed assets	557	557	557	557
Deferred assets	131	131	131	131
Other assets	172	172	172	172
Total Assets	**5,048**	**5,350**	**5,691**	**6,124**
S/T borrowings	0	0	0	0
AP/NP	188	208	268	325
Other ST liabilities	229	229	229	229
LT debt	0	0	0	0
Other LT liabilities	141	141	141	141
Total Liabilities	**558**	**579**	**638**	**695**
Common shares	167	167	167	167
Additional capital	3,164	3,164	3,164	3,164
Retained earning	1,178	1,459	1,741	2,116
Other shareholders' equity	(18)	(18)	(18)	(18)
Total Equity	**4,490**	**4,771**	**5,053**	**5,428**
Total Liab. & Shrhldr's Equity	**5,048**	**5,350**	**5,691**	**6,124**

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Cash Flow Statement Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**523**	**479**	**454**	**553**
Net profits	498	538	667	760
Depreciation	24	24	24	24
Working Capital Change	(126)	(83)	(236)	(230)
Other adjustments	127	0	0	0
Cashflow from Investing	**(1,913)**	**(24)**	**(24)**	**(24)**
Capex	(516)	(24)	(24)	(24)
Change of LT Investment	(1,972)	0	0	0
Change of ST Investment	0	0	0	0
Other adjustments	575	0	0	0
Cashflow from financing	**1,757**	**(257)**	**(385)**	**(385)**
Increase in L/T debt	0	0	0	0
Increase in S/T debt	(1)	0	0	0
Cash Dividend Paid	(71)	(257)	(385)	(385)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	1,842	0	0	0
Other adjustments	(13)	0	0	0
Exchange rate adjustment	(16)	0	0	0
Net change in cash	**351**	**198**	**46**	**145**

Financial Ratios	2025	2026E	2027E	2028E
Growth(%)				
Turnover	27.8	19.0	24.6	19.7
Operating profits	63.2	19.1	27.0	15.4
EPS	46.7	8.1	24.0	14.0
Margins (%)				
Gross Margin	46.6	46.3	44.6	43.9
Operating Margin	16.5	16.5	16.8	16.2
Pretax Margin	19.8	18.4	18.4	17.5
Net Margin	19.4	17.6	17.5	16.7
Return (%)				
ROAE	14.9	11.6	13.6	14.5
ROAA	12.9	10.3	12.1	12.9
Gearing (%)				
Net Debt/Equity	(22.8)	(25.6)	(25.1)	(26.1)
Liabilities/Equity	12.4	12.1	12.6	12.8
Ratios (X)				
Current ratio	9.4	9.7	9.2	9.0
Quick ratio	3.3	3.8	3.6	3.7
Others				
AR/NR Turnover (days)	51	51	51	51
Inventory Turnover (days)	148	148	148	148
AP Turnover (days)	46	46	46	46
Cash Conversion (days)	153	153	153	153

Source: Company data, Morgan Stanley Research (E) estimates

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Nuvoton: Earnings Estimate Revisions

We factor in the actual 4Q25 results and 1Q26 preliminary revenue, expect a loss in 2026, raise our 2027 EPS by 12%, and introduce our 2028 forecast: We expect Nuvoton's utilization to improve in 2026 vs. 2025, but at a milder pace; we therefore slightly lower our sales forecast for 2027. We also expect Nuvoton to face some margin pressure in 2026 from higher material costs, although this should be modest, and therefore lower our margin assumptions. We expect margin improvement in 2027E, supported by new product introductions and spec migration.

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Exhibit 11: Nuvoton: Estimate revisions

NT$ mn	New 2026E	Previous 2026E	Diff.	New 2027E	Previous 2027E	Diff.	New 2028E
Net sales	32,309	32,087	1%	36,825	38,437	-4%	40,494
COGS	(20,683)	(19,515)		(22,651)	(23,035)		(23,854)
Gross profit	11,626	12,572	-8%	14,174	15,403	-8%	16,640
Operating expenses	(12,406)	(12,569)		(12,747)	(14,232)		(13,389)
Operating profit	(781)	3	n.a.	1,427	1,171	22%	3,252
Non-op. income (exp.)	177	409		726	756		600
Pretax Income	(603)	412	n.a.	2,153	1,926	12%	3,852
Taxes	84	(58)		(301)	(270)		(539)
Net income	(519)	354	n.a.	1,852	1,657	12%	3,312
Reported EPS (NT$)	(1.24)	0.84	**n.a.**	4.41	3.94	**12%**	7.89
Margins							
Gross margin	36.0%	39.2%	-3.2 ppt	38.5%	40.1%	-1.6 ppt	41.1%
Operating margin	-2.4%	0.0%	-2.4 ppt	3.9%	3.0%	0.8 ppt	8.0%
Pretax margin	-1.9%	1.3%	-3.2 ppt	5.8%	5.0%	0.8 ppt	9.5%
Net margin	-1.6%	1.1%	-2.7 ppt	5.0%	4.3%	0.7 ppt	8.2%

Source: Company data, Morgan Stanley Research estimates

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Exhibit 12: Nuvoton: Quarterly Financials

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26e	2Q26e	3Q26e	4Q26e	2025	2026e	2027e	2028e
Total Revenues	**8,364**	**7,816**	**7,272**	**7,040**	**7,988**	**8,319**	**8,057**	**7,945**	**30,492**	**32,309**	**36,825**	**40,494**
Sequential Change	*15.1%*	*-6.6%*	*-7.0%*	*-3.2%*	*13.5%*	*4.2%*	*-3.1%*	*-1.4%*				
Change vs Year Ago	*-1.2%*	*-5.0%*	*-8.6%*	*-3.2%*	*-4.5%*	*6.4%*	*10.8%*	*12.8%*	*-4.5%*	*6.0%*	*14.0%*	*10.0%*
Cost of Sales	**(4,914)**	**(5,088)**	**(4,675)**	**(4,627)**	**(5,202)**	**(5,305)**	**(5,127)**	**(5,049)**	**(19,304)**	**(20,683)**	**(22,651)**	**(23,854)**
Percent of Revenues	*58.7%*	*65.1%*	*64.3%*	*65.7%*	*65.1%*	*63.8%*	*63.6%*	*63.5%*	*63.3%*	*64.0%*	*61.5%*	*58.9%*
Gross Profit	**3,451**	**2,728**	**2,597**	**2,413**	**2,785**	**3,014**	**2,931**	**2,896**	**11,189**	**11,626**	**14,174**	**16,640**
Gross Margin	41.3%	34.9%	35.7%	34.3%	34.9%	36.2%	36.4%	36.5%	36.7%	36.0%	38.5%	41.1%
Total Opex	(3,177)	(3,030)	(3,113)	(3,097)	(3,244)	(3,162)	(3,030)	(2,970)	(12,417)	(12,406)	(12,747)	(13,389)
Q/Q	*6.3%*	*-4.7%*	*2.8%*	*-0.5%*	*4.8%*	*-2.5%*	*-4.2%*	*-2.0%*				
Y/Y	*3.2%*	*0.6%*	*3.6%*	*3.6%*	*2.1%*	*4.4%*	*-2.7%*	*-4.1%*	*2.8%*	*-0.1%*	*2.7%*	*5.0%*
Percent of Revenues	*38.0%*	*38.8%*	*42.8%*	*44.0%*	*40.6%*	*38.0%*	*37.6%*	*37.4%*	*40.7%*	*38.4%*	*34.6%*	*33.1%*
R&D	(2,303)	(2,196)	(2,293)	(2,257)	(2,396)	(2,329)	(2,216)	(2,169)	(9,049)	(9,110)	(9,482)	(9,767)
Percent of Revenues	*27.5%*	*28.1%*	*31.5%*	*32.1%*	*30.0%*	*28.0%*	*27.5%*	*27.3%*	*29.7%*	*28.2%*	*25.7%*	*24.1%*
Sales and Marketing	(254)	(250)	(249)	(254)	(272)	(266)	(258)	(254)	(1,007)	(1,050)	(1,179)	(1,296)
Percent of Revenues	*3.0%*	*3.2%*	*3.4%*	*3.6%*	*3.4%*	*3.2%*	*3.2%*	*3.2%*	*3.3%*	*3.3%*	*3.2%*	*3.2%*
General and Admin	(620)	(584)	(571)	(586)	(576)	(566)	(556)	(546)	(2,361)	(2,246)	(2,086)	(2,326)
Percent of Revenues	*7.4%*	*7.5%*	*7.8%*	*8.3%*	*7.2%*	*6.8%*	*6.9%*	*6.9%*	*7.7%*	*7.0%*	*5.7%*	*5.7%*
Operating Income	**273**	**(302)**	**(516)**	**(684)**	**(459)**	**(148)**	**(99)**	**(74)**	**(1,228)**	**(781)**	**1,427**	**3,252**
Percent of Revenues	*3.3%*	*-3.9%*	*-7.1%*	*-9.7%*	*-5.7%*	*-1.8%*	*-1.2%*	*-0.9%*	*-4.0%*	*-2.4%*	*3.9%*	*8.0%*
% QoQ	*-224.2%*	*-210.4%*	*71.1%*	*32.4%*	*-32.8%*	*-67.8%*	*-32.8%*	*-25.8%*	**(1,228)**	**(781)**	**1,427**	**3,252**
Total Non-operating Income(Loss)	**59**	**(343)**	**45**	**22**	**76**	**63**	**0**	**38**	**(217)**	**177**	**726**	**600**
Forex gain (loss)	35	(588)	0	0	0	0	0	0	(553)	0	0	0
Interest expense	(40)	(52)	(50)	(41)	0	0	0	0	(182)	0	0	0
Interest income	72	60	43	37	76	63	0	38	212	177	276	240
Net Investment Income	(14)	155	0	0	0	0	0	0	141	0	450	360
Investment Disposal Gain (loss)	0	0	0	0	0	0	0	0	0	0	0	0
Other income (expense)	6	82	52	26	0	0	0	0	166	0	0	0
Profit Before Taxes	**333**	**(644)**	**(471)**	**(662)**	**(383)**	**(85)**	**(99)**	**(35)**	**(1,445)**	**(603)**	**2,153**	**3,852**
Percent of Revenues	*4.0%*	*-8.2%*	*-6.5%*	*-9.4%*	*-4.8%*	*-1.0%*	*-1.2%*	*-0.4%*	*-4.7%*	*-1.9%*	*5.8%*	*9.5%*
Change vs Year Ago	*-17.9%*	*-1410.6%*	*1227.5%*	*3329.9%*	*-215.2%*	*-86.8%*	*-78.9%*	*-94.6%*				
Taxes	(115)	(30)	(43)	(32)	54	12	14	5	(220)	84	(301)	(539)
Tax Rate	*34.6%*	*-4.7%*	*-9.0%*	*-4.8%*	*14.0%*	*14.0%*	*14.0%*	*14.0%*	*-15.2%*	*14.0%*	*14.0%*	*14.0%*
Reported Income (TW GAAP)	**217**	**(675)**	**(513)**	**(694)**	**(330)**	**(73)**	**(86)**	**(30)**	**(1,665)**	**(519)**	**1,852**	**3,312**
Percent of Revenues	*2.6%*	*-8.6%*	*-7.1%*	*-9.9%*	*-4.1%*	*-0.9%*	*-1.1%*	*-0.4%*	*-5.5%*	*-1.6%*	*5.0%*	*8.2%*
Change vs Year Ago	*-46.8%*	*3061.4%*	*386.9%*	*1011.9%*	*-251.6%*	*-89.1%*	*-83.3%*	*-95.6%*	*-857.2%*	*-68.8%*	*-456.8%*	*78.9%*
MW Diluted EPS (NT$, TW GAAP)	**0.52**	**(1.61)**	**(1.22)**	**(1.65)**	**(0.79)**	**(0.17)**	**(0.20)**	**(0.07)**	**(3.96)**	**(1.24)**	**4.41**	**7.89**
EBS Dilution (%)	*11.9%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*	*11.9%*	*11.9%*	*11.9%*	*11.9%*
Reported Basic EPS (NT$, GAAP)	**0.52**	**(1.61)**	**(1.22)**	**(1.65)**	**(0.79)**	**(0.17)**	**(0.20)**	**(0.07)**	**(3.96)**	**(1.24)**	**4.41**	**7.89**
Sequential Change	*-448.4%*	*-410.3%*	*-24.1%*	*35.5%*	*-52.5%*	*-77.8%*	*16.7%*	*-64.4%*	*-856.7%*	*-68.8%*	*-456.8%*	*78.9%*

Source: Company data, Morgan Stanley Research

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Nuvoton: Valuation Methodology

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We raise our PT from NT$48 to NT$98: We factor in our earnings estimate changes for 2026-2027e. The increase in target price is driven mainly by our newly introduced 2028 forecast, which affects the medium-term outlook, while we keep the other RI model assumptions unchanged, including a 9.2% cost of equity (beta of 1.2, risk-free rate of 2.0%, and risk premium of 6.0%), payout ratio of 55%, a medium-term growth rate of 7.0%, and a terminal growth rate of 3.5%. Our bull and bear case scenario values are also raised from NT$68 and NT$23 to NT$139 and NT$47, implying 31x and 11x 2027e EPS.

Exhibit 13: Nuvoton: RI model

(NT$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	**12,609**	**14,461**	**17,773**	**19,563**	**21,478**	**23,528**	**25,720**	**28,067**	**30,577**	**33,263**	**36,137**	**39,213**
Net Profit	**(519)**	**1,852**	**3,312**	**3,544**	**3,792**	**4,058**	**4,342**	**4,646**	**4,971**	**5,319**	**5,691**	**6,090**
ROAE	**NM**	**13.7%**	**20.6%**	**19.0%**	**18.5%**	**18.0%**	**17.6%**	**17.3%**	**17.0%**	**16.7%**	**16.4%**	**16.2%**
Residual Income	NA	565	1,642	1,739	1,816	1,897	1,984	2,077	2,176	2,282	2,396	2,517
Spread	**NA**	**4.5%**	**11.4%**	**9.8%**	**9.3%**	**8.8%**	**8.4%**	**8.1%**	**7.8%**	**7.5%**	**7.2%**	**7.0%**
2026	1.00	0.92	0.84	0.77	0.70	0.64	0.59	0.54	0.49	0.45	0.41	0.38
Ending Equity Capital	12,609											
PV of Forecast Period	11,124											
PV of Continuing Value	17,356											
Equity Value	**41,088**											
No. of Shares	420											
Projected Price (NT$)	**98**											

Source: Company data, Morgan Stanley Research estimates

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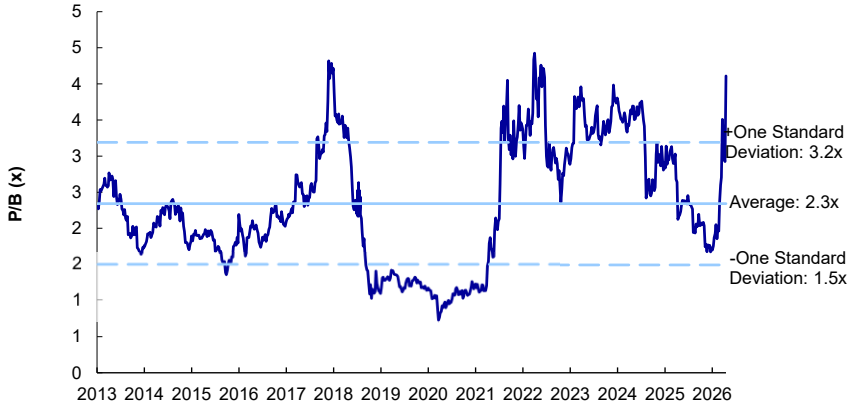
Exhibit 14: Nuvoton: P/B



Source: Company data, Morgan Stanley Research estimates

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Risk Reward – Nuvoton Technology Corporation (4919.TW)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Gross margin (%)	37	36	38	41
MCU ASP growth (%)	(5)	(1)	(5)	(6)
Operating margin (%)	(5)	(2)	5	8

INVESTMENT DRIVERS

- MCU unit shipments
- MCU pricing trend
- Operating expenses
- Automotive demand

GLOBAL REVENUE EXPOSURE

- 0-10% Europe ex UK
- 0-10% Japan
- 0-10% North America
- 20-30% APAC, ex Japan, Mainland China and India
- 50-60% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

- MCU up-cycle is more sustained than expected.
- China's MCU localization is faster than expected.
- Traction from auto and BMC business is greater.
- Margin improvement is more significant.

RISKS TO DOWNSIDE

- MCU up-cycle ends earlier with severe pricing erosion.
- China's MCU localization is slower than expected.
- Traction from auto and BMC business is less.
- Margins contract amid more intensified competition.

OWNERSHIP POSITIONING

Inst. Owners, % Active 19%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)
32,309
32,087 — 33,542
32,649

EBIT (NT$, mn)
(781)
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (NT$)
(1.24)
Note: There are not sufficient brokers supplying consensus data for this metric

ROE (%)
(4.0)
(10.9) — 2.5
(2.4)

Mean Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Nuvoton: Financial Summary

Exhibit 15: Nuvoton: Financial Summary

Income Statement, 2024-2027e, Year End Dec

(NT$ mn)	2025	2026e	2027e	2028e
Turnover	**30,492**	**32,309**	**36,825**	**40,494**
YoY Growth	*-4.5%*	*6.0%*	*14.0%*	*10.0%*
Less: COGS	**(19,304)**	**(20,683)**	**(22,651)**	**(23,854)**
Variable costs	-	-	-	-
Depreciation & amort	-	-	-	-
Gross profit	**11,189**	**11,626**	**14,174**	**16,640**
YoY Growth	*-7.5%*	*3.9%*	*21.9%*	*17.4%*
% margin	*36.7%*	*36.0%*	*38.5%*	*41.1%*
Operating Expenses:	**(12,417)**	**(12,406)**	**(12,747)**	**(13,389)**
R&D	(9,049)	(9,110)	(9,482)	(9,767)
Sales and Marketing	(1,007)	(1,050)	(1,179)	(1,296)
General and Admin	(2,361)	(2,246)	(2,086)	(2,326)
Operating Profit	**(1,228)**	**(781)**	**1,427**	**3,252**
YoY Growth	*-9669.4%*	*-36.4%*	*-282.9%*	*127.8%*
% margin	*-4.0%*	*-2.4%*	*3.9%*	*8.0%*
Total Non-op	**(217)**	**177**	**726**	**600**
Pretax Profit	(1,445)	(603)	2,153	3,852
YoY Growth	*-461.5%*	*-58.2%*	*-456.8%*	*78.9%*
% margin	*-4.7%*	*-1.9%*	*5.8%*	*9.5%*
Tax	(220)	84	(301)	(539)
Net Income bf. Emp. Bonus	(1,889)	(589)	2,102	3,760
Employee bonus expense	225	70	(250)	(447)
Reported Net Income	(1,665)	(519)	1,852	3,312
Reported EPS (NT$)	**(3.96)**	**(1.24)**	**4.41**	**7.89**
ModelWare EPS (NT$)	**(3.96)**	**(1.24)**	**4.41**	**7.89**

Balance sheet

(NT$ mn)	2025	2026e	2027e	2028e
Assets				
Cash & Equivalents	7,203	4,962	5,049	9,786
Marketable Security	1	1	1	1
A/R & N/R	3,964	4,473	5,831	5,847
Inventories	6,217	6,783	8,421	8,065
Other Current Asset	917	917	917	917
Total current assets	**18,301**	**17,136**	**20,219**	**24,616**
LT Investment	863	863	1,313	1,673
Total Fixed Assets	5,836	6,675	7,568	7,999
Total Other Assets	4,992	4,992	4,992	4,992
Total Assets	**29,991**	**29,665**	**34,091**	**39,279**
A/P & N/P	2,187	2,379	2,954	2,829
Accrued Expenses	13	13	13	13
Other Payable	3,430	3,430	3,430	3,430
Total Current Liab.	**8,842**	**9,035**	**9,610**	**9,485**
L-T Liabilities	4,865	4,865	6,865	8,865
Total Other L-T Liab	3,156	3,156	3,156	3,156
Total Liabilities	**16,874**	**17,056**	**19,630**	**21,506**
Common Stock	4,198	4,198	4,198	4,198
Preferred Stock				
Capital Reserve	7,100	7,100	7,100	7,100
Retained earnings	1,820	1,301	3,153	6,466
Treasury Stock	0	10	10	10
Total Equity	**13,118**	**12,609**	**14,461**	**17,773**
Total Liab. & Equity				

Key Ratios

	2025	2026e	2027e	2028e
Return (%)				
ROAA	*(5.1%)*	*(1.6%)*	*5.4%*	*9.0%*
ROAE	*(11.3%)*	*(3.6%)*	*12.4%*	*20.6%*
OP. ATO	*164.6%*	*157.5%*	*147.6%*	*155.3%*
Gearing (x)				
Net Debt/ Equity	(0.2x)	(0.0x)	0.1x	(0.1x)
Current Ratio	2.1x	1.9x	2.1x	2.6x
Quick Ratio	1.3x	1.0x	1.1x	1.6x
Operating Cycle				
AR/NR Turnover (days)	56	56	53	52
Inventory Turnover (days)	137	128	123	124
AP Turnover (days)	48	45	43	44
Cash Conversion (days)	145	139	133	133

Cash Flow Statement

(NT$ mn)	2025	2026e	2027e	2028e
Net Income	(1,665)	(519)	1,852	3,312
Depreciation	1,767	196	201	664
Net Investment Losses (Gains)	(0)	0	0	0
Others	(392)	(883)	(2,871)	(145)
Cash Flow-Operating	**(290)**	**(1,206)**	**(818)**	**3,831**
(Purchase) of FA	(986)	(1,035)	(1,094)	(1,165)
Sale of Fix Asset	3	0	0	0
(Purchase)L-T Inv.	0	0	0	0
Sale of L-T Inv.	0	0	0	0
Sale(Pur.)S-T Inv.	54	0	0	0
Cash Flow-Investing	**(1,211)**	**(1,035)**	**(1,094)**	**(1,165)**
Inc(Dec)-S-T Debt	(857)	0	2,000	2,000
Dividend Paid	(168)	0	0	0
Dir.&Emp.Bonus	0	0	0	0
Proceed from New Issue	0	0	0	0
Dec/Inc-TreasureSt	0	0	0	0
Others	4,348	0	0	0
Cash Flow-Financing	**3,323**	**0**	**2,000**	**2,000**
Change in Cash	**1,499**	**(2,241)**	**88**	**4,666**
Net cash/(debt), b/f	5,704	7,203	4,962	5,049
Net cash/(debt), c/f	**7,203**	**4,962**	**5,049**	**9,715**

Source: Company data, Morgan Stanley Research estimates

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GigaDevice Semiconductor Beijing Inc (603986.SS)

Our price target of Rmb301 is our base case value, derived from our residual income model. Key assumptions: cost of equity of 8.9% (beta 1.1, risk-free rate 2% and risk premium 6%), a payout ratio of 40%, medium-term growth rate of 16.0%, and a terminal growth rate of 3.0%.

Risks to Upside

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- NOR up-cycle driven by stronger demand
- Superior chip design, leading to increasing exposure to low-density NOR
- Faster-than-expected DRAM development

Risks to Downside

- NOR down-cycle driven by weaker demand
- Inferior chip design, leading to increasing exposure to mid-/high-density NOR
- Slower-than-expected DRAM development

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Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

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4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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